

Commerce XI Maths App

Activities 1.1–1.10

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 1 - Sets and Relations | Total questions: 23

Question 1 - A1.1

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3\}$, and $B = \{3, 4\}$.

Find $(A \cup B)'$.

Options

A. $\{5, 6, 7, 8, 9, 10\}$

B. $\{1, 2, 3, 4\}$

C. $\{3\}$

D. $\{1, 2, 4, 5, 6, 7, 8, 9, 10\}$

Correct answer: $\{5, 6, 7, 8, 9, 10\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{5, 6, 7, 8, 9, 10\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 2 - A1.2

Let A and B be nonempty subsets of a universal set.

Which condition makes $A \cup B = A \cap B$?

Options

A. $A = B$

B. $A \cap B = \{ \}$

C. $A \subset B$ only

D. $B \subset A$ only

Correct answer: $A = B$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A = B$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 3 - A1.3

Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Which statement is correct?

Options

A. $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$

B. $A \times B = B \times A$, but $n(A \times B) \neq n(B \times A)$

C. $A \times B = A \cup B$

D. $n(A \times B) = n(A) + n(B)$

Correct answer: $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A \times B \neq B \times A$, but $n(A \times B) = n(B \times A)$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 4 - A1.4

What conclusion will you draw about two sets A and B if $A \subseteq B$ and $B \subseteq A$?

Options

A. $A = B$

B. $A \cap B = \{ \}$

C. A is a proper subset of B

D. B is a proper subset of A

Correct answer: $A = B$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A = B$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 5 - A1.5

If $A = \{-1/2, 2/3\}$, which set builder form represents A?

Options

A. $\{x / 6x^2 - x - 2 = 0\}$

B. $\{x / 6x^2 + x - 2 = 0\}$

C. $\{x / 6x^2 - x + 2 = 0\}$

D. $\{x / 3x^2 - x - 2 = 0\}$

Correct answer: $\{x / 6x^2 - x - 2 = 0\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / 6x^2 - x - 2 = 0\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 6 - A1.6 Domain

For the relation $R = \{(1, 5), (2, 6), (3, 7), (4, 8)\}$, write the domain.

Options

- A. {1, 2, 3, 4}
- B. {5, 6, 7, 8}
- C. {1, 2, 3, 4, 5, 6, 7, 8}
- D. {1, 5}

Correct answer: {1, 2, 3, 4}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {1, 2, 3, 4}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 7 - A1.6 Range

For the relation $R = \{(1, 5), (2, 6), (3, 7), (4, 8)\}$, write the range.

Options

- A. {5, 6, 7, 8}
- B. {1, 2, 3, 4}
- C. {1, 2, 3, 4, 5, 6, 7, 8}
- D. {4, 8}

Correct answer: {5, 6, 7, 8}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {5, 6, 7, 8}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 8 - A1.7(i)

Write the following set in Roster form.

A – set of all factors of 24.

Options

- A. $A = \{1, 2, 3, 4, 6, 8, 12, 24\}$
- B. $A = \{1, 2, 3, 4, 6, 8, 12\}$
- C. $A = \{2, 3, 4, 6, 8, 12, 24\}$
- D. $A = \{1, 2, 3, 4, 6, 8, 12, 24, 48\}$

Correct answer: $A = \{1, 2, 3, 4, 6, 8, 12, 24\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A = \{1, 2, 3, 4, 6, 8, 12, 24\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 9 - A1.7(ii)

Write the following set in Roster form.

B – set of all prime numbers less than 30.

Options

A. $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

B. $B = \{1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

C. $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$

D. $B = \{2, 3, 5, 7, 9, 11, 13, 17, 19, 23, 29\}$

Correct answer: $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $B = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 10 - A1.7(iii)

Write the following set in Roster form.

C – set of all letters in the word “MATHEMATICS”.

Options

A. $C = \{M, A, T, H, E, I, C, S\}$

B. $C = \{M, A, T, H, E, I, C\}$

C. $C = \{M, A, T, H, E, I, C, S, M\}$

D. $C = \{M, A, T, H, E, I, S\}$

Correct answer: $C = \{M, A, T, H, E, I, C, S\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $C = \{M, A, T, H, E, I, C, S\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 11 - A1.8

In a survey of 400 students, it was found that 150 drink milk, 250 drink Tea, and 50 drink both.

How many drink neither Tea nor milk?

Correct answer: 50

Explanation and steps

Use $n(M \cup T) = 150 + 250 - 50 = 350$. Therefore, neither Tea nor milk = $400 - 350 = 50$.

Question 12 - A1.9 Union

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $A \cup B$.

Options

A. $\{1/2, 1/3, 1/4, 1/6, 1/8, 1/9, 1/10, 1/12, 1/14, 1/15, 1/18, 1/21\}$

B. $\{1/6, 1/12\}$

C. $\{1/3, 1/9, 1/15, 1/18, 1/21\}$

D. $\{1/2, 1/4, 1/8, 1/10, 1/14\}$

Correct answer: $\{1/2, 1/3, 1/4, 1/6, 1/8, 1/9, 1/10, 1/12, 1/14, 1/15, 1/18, 1/21\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1/2, 1/3, 1/4, 1/6, 1/8, 1/9, 1/10, 1/12, 1/14, 1/15, 1/18, 1/21\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 13 - A1.9 Intersection

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $A \cap B$.

Options

A. $\{1/6, 1/12\}$

B. $\{1/3, 1/6, 1/9, 1/12\}$

C. $\{1/2, 1/4, 1/6, 1/8\}$

D. $\{\}$

Correct answer: $\{1/6, 1/12\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1/6, 1/12\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 14 - A1.9 A – B

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $A - B$.

Options

A. $\{1/3, 1/9, 1/15, 1/18, 1/21\}$

B. $\{1/2, 1/4, 1/8, 1/10, 1/14\}$

C. $\{1/6, 1/12\}$

D. $\{1/3, 1/6, 1/9, 1/12, 1/15, 1/18, 1/21\}$

Correct answer: $\{1/3, 1/9, 1/15, 1/18, 1/21\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1/3, 1/9, 1/15, 1/18, 1/21\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 15 - A1.9 B – A

$$A = \{1/(3x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

$$B = \{1/(2x) \mid x \in \mathbb{N} \text{ and } x < 8\}$$

Find $B - A$.

Options

A. $\{1/2, 1/4, 1/8, 1/10, 1/14\}$

B. $\{1/3, 1/9, 1/15, 1/18, 1/21\}$

C. $\{1/6, 1/12\}$

D. $\{1/2, 1/4, 1/6, 1/8, 1/10, 1/12, 1/14\}$

Correct answer: $\{1/2, 1/4, 1/8, 1/10, 1/14\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1/2, 1/4, 1/8, 1/10, 1/14\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 16 - A1.10 Complements

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{4, 5, 6, 7, 8\}.$$

Find A' and B' .

Options

A. $A' = \{6, 7, 8\}, B' = \{1, 2, 3\}$

$$B. A' = \{1, 2, 3\}, B' = \{6, 7, 8\}$$

$$C. A' = \{4, 5\}, B' = \{4, 5\}$$

$$D. A' = \{6, 7, 8\}, B' = \{1, 2, 3, 4, 5\}$$

Correct answer: $A' = \{6, 7, 8\}, B' = \{1, 2, 3\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A' = \{6, 7, 8\}, B' = \{1, 2, 3\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 17 - A1.10 Intersection

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{4, 5, 6, 7, 8\}.$$

Find $A \cap B$ and $n(A \cap B)$.

Options

$$A. A \cap B = \{4, 5\}, n(A \cap B) = 2$$

$$B. A \cap B = \{1, 2, 3\}, n(A \cap B) = 3$$

$$C. A \cap B = \{6, 7, 8\}, n(A \cap B) = 3$$

$$D. A \cap B = \{1, 2, 3, 4, 5, 6, 7, 8\}, n(A \cap B) = 8$$

Correct answer: $A \cap B = \{4, 5\}, n(A \cap B) = 2$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A \cap B = \{4, 5\}, n(A \cap B) = 2$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 18 - A1.10 A – B

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{4, 5, 6, 7, 8\}.$$

Find $A - B$ and $n(A - B)$.

Options

$$A. A - B = \{1, 2, 3\}, n(A - B) = 3$$

$$B. A - B = \{4, 5\}, n(A - B) = 2$$

$$C. A - B = \{6, 7, 8\}, n(A - B) = 3$$

$$D. A - B = \{1, 2, 3, 4, 5\}, n(A - B) = 5$$

Correct answer: $A - B = \{1, 2, 3\}, n(A - B) = 3$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A - B = \{1, 2, 3\}$, $n(A - B) = 3$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 19 - A1.10 B – A

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{4, 5, 6, 7, 8\}.$$

Find $B - A$ and $n(B - A)$.

Options

A. $B - A = \{6, 7, 8\}$, $n(B - A) = 3$

B. $B - A = \{1, 2, 3\}$, $n(B - A) = 3$

C. $B - A = \{4, 5\}$, $n(B - A) = 2$

D. $B - A = \{4, 5, 6, 7, 8\}$, $n(B - A) = 5$

Correct answer: $B - A = \{6, 7, 8\}$, $n(B - A) = 3$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $B - A = \{6, 7, 8\}$, $n(B - A) = 3$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 20 - A1.10 Union

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{4, 5, 6, 7, 8\}.$$

Find $A \cup B$ and $n(A \cup B)$.

Options

A. $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $n(A \cup B) = 8$

B. $A \cup B = \{4, 5\}$, $n(A \cup B) = 2$

C. $A \cup B = \{1, 2, 3, 6, 7, 8\}$, $n(A \cup B) = 6$

D. $A \cup B = \{1, 2, 3, 4, 5\}$, $n(A \cup B) = 5$

Correct answer: $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $n(A \cup B) = 8$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $n(A \cup B) = 8$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 21 - A1.10(i)

If $n(A - B) = 3$, $n(A \cap B) = 2$, and $n(B - A) = 3$, find $n(A - B) + n(A \cap B) + n(B - A)$.

Options

- A. 8
- B. 5
- C. 6
- D. 9

Correct answer: 8

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 8. The other choices omit a required element, include an invalid element, or use a different operation.

Question 22 - A1.10(ii)

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$.

Complete $A \cap B' = A - B$.

Options

- A. $A \cap B' = \{1, 2, 3\}$
- B. $A \cap B' = \{4, 5\}$
- C. $A \cap B' = \{6, 7, 8\}$
- D. $A \cap B' = \{1, 2, 3, 4, 5\}$

Correct answer: $A \cap B' = \{1, 2, 3\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A \cap B' = \{1, 2, 3\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 23 - A1.10(iii)

$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$.

Complete $A' \cap B = B - A$.

Options

- A. $A' \cap B = \{6, 7, 8\}$
- B. $A' \cap B = \{1, 2, 3\}$
- C. $A' \cap B = \{4, 5\}$

D. $A' \cap B = \{4, 5, 6, 7, 8\}$

Correct answer: $A' \cap B = \{6, 7, 8\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $A' \cap B = \{6, 7, 8\}$. The other choices omit a required element, include an invalid element, or use a different operation.